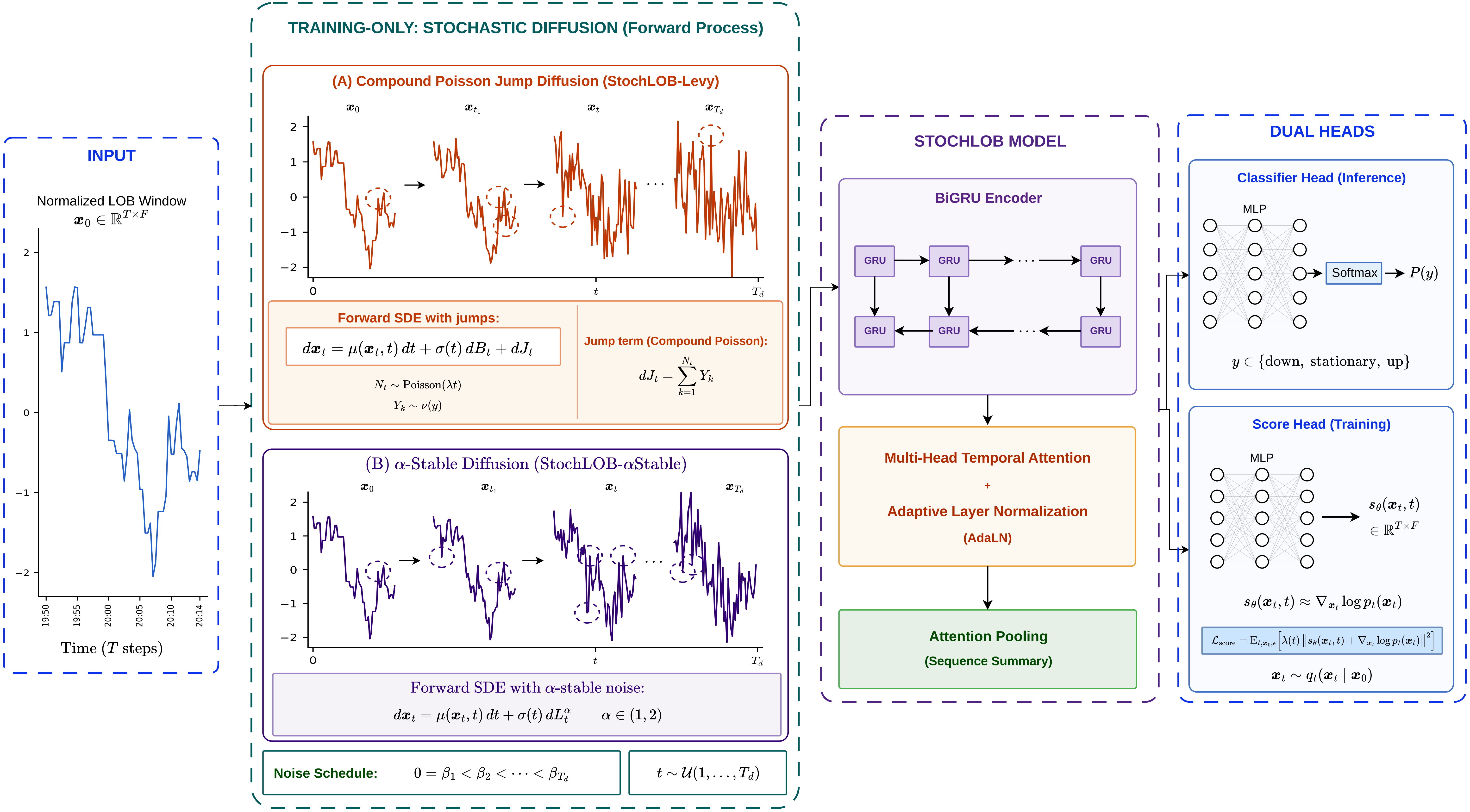

INPUT
Normalized LOB Window
$x_0 \in \mathbb{R}^{T \times F}$
Time (T steps)
TRAINING-ONLY: STOCHASTIC DIFFUSION (Forward Process)
(A) Compound Poisson Jump Diffusion (StochLOB-Levy)
Forward SDE with jumps:
$dx_t = \mu(x_t, t)\,dt + \sigma(t)\,dB_t + dJ_t$
$N_t \sim \text{Poisson}(\lambda t)$
$Y_k \sim \nu(y)$
Jump term (Compound Poisson):
$dJ_t = \sum_{k=1}^{N_t} Y_k$
(B) α-Stable Diffusion (StochLOB-αStable)
Forward SDE with α-stable noise:
$dx_t = \mu(x_t, t)\,dt + \sigma(t)\,dL_t^{\alpha} \quad \alpha \in (1, 2)$
Noise Schedule: $0 = \beta_1 < \beta_2 < \cdots < \beta_{T_d}$
$t \sim \mathcal{U}(1, \ldots, T_d)$
STOCHLOB MODEL
BiGRU Encoder
GRU
Multi-Head Temporal Attention
+
Adaptive Layer Normalization
(AdaLN)
Attention Pooling
(Sequence Summary)
DUAL HEADS
Classifier Head (Inference)
MLP
Softmax
$P(y)$
$y \in \{\text{down, stationary, up}\}$
Score Head (Training)
MLP
$s_\theta(x_t, t)$
$\in \mathbb{R}^{T \times F}$
$s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t)$
$\mathcal{L}_{\text{score}} = \mathbb{E}_{t, x_0, \epsilon}\left[\lambda(t)\left\| s_\theta(x_t, t) + \nabla_{x_t} \log p_t(x_t)\right\|^2\right]$
$x_t \sim q_t(x_t \mid x_0)$